UNIT 3 · TOPIC 12(C)

Batteries

Syllabus topic: Measurement of engine power; electric motors and their characteristics including batteries

Lead-acid (SLI and deep-cycle)

- **2.1 V/cell** (6 cells → 12 V); electrolyte H_2SO_4.
- Specific gravity ~**1.26–1.28** fully charged, ~**1.12–1.15** discharged — check the state of charge with a **hydrometer**.
- Capacity in **ampere-hours** at a rated discharge (e.g., C20).
- Sulphation develops from standing discharged; top up with distilled water.
- Energy density only **30–40 Wh/kg**.

Lithium-ion / LiPo

- **3.6–3.7 V/cell**; **150–250 Wh/kg**; high cycle life.
- Needs a **Battery Management System (BMS)** — cell balancing, over-charge/discharge and thermal protection.
- Powers e-tractors, battery sprayers and drones (Unit 2, Topic 12).

$$\text{Energy (Wh)} = \text{V} \times \text{Ah}$$

Example. 12 V × 100 Ah = **1.2 kWh** — runs a 600 W load ~**2 h** (before depth-of-discharge limits).

Practice MCQs

Q166. Each cell of a lead-acid battery gives about:

(a) 1.1 V (b) 1.5 V
(c) 2.1 V (d) 3.7 V

Q167. Specific gravity of electrolyte in a FULLY CHARGED lead-acid battery is about:

(a) 1.00 (b) 1.13
(c) 1.26–1.28 (d) 1.45

Q168. Battery state of charge in the field is checked with a:

(a) Hydrometer (b) Anemometer
(c) Manometer (d) Planimeter

Q169. Battery capacity is rated in:

(a) Watts (b) Ampere-hours
(c) Joules per kg only (d) Volts

Q170. Energy stored in a 12-V, 100-Ah battery is:

(a) 0.12 kWh (b) 1.2 kWh
(c) 12 kWh (d) 120 kWh

Q171. Compared with lead-acid, lithium-ion batteries have:

(a) Lower energy density

(b) 4–6× higher energy density (Wh/kg) with BMS protection

(c) Same density

(d) No cycle life

Answers

Q166 — (c) 2.1 V

Q167 — (c) 1.26–1.28

Q168 — (a) Hydrometer

Q169 — (b) Ampere-hours

Q170 — (b) 1.2 kWh — *E = V×Ah = 12×100 = 1200 Wh = 1.2 kWh.*

Q171 — (b) 4–6× higher energy density (Wh/kg) with BMS protection

Sources: Tractors and their Power Units (Liljedahl et al.); Kepner et al.; BIS publications; ISO/OECD/ASABE documentation; NPTEL lectures on farm power and IC engines.